Fiscal Year:	FY 2025	**Task Last Updated:**	FY 06/04/2024
PI Name:	Komatsu, David Ph.D.		
Project Title:	Bone Remodeling under Differential Gravitational Environments		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	11794-8181	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	01/31/2019	**End Date:**	01/30/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:	NOTE: Start date changed to 01/31/2019 per F. Hernandez/ARC (Ed., 6/30/24). NOTE: End date changed to 01/30/2024 per NSSC information (Ed., 4/12/23). NOTE: End date changed to 01/30/2022 per F. Hernandez/ARC (Ed., 11/4/21). NOTE: End date changed to 02/28/2022 per NSSC (Ed., 11/4/21).		
Key Personnel Changes/Previous PI:	Kaitlin Farrell was added as a research technician in January, 2022. Isaiah Taylor was added as a graduate student in January, 2022. NOTE: December 2021: Dr. Sardar Uddin is still consulting on the current award as a CoInvestigator; Dr. David Komatsu serves as Principal Investigator (PI). NOTE March 2020: New PI is Dr. David Komatsu, who replaces previous PI - Dr. Sardar Uddin; Dr. Uddin remains on the project as a CoInvestigator.		
COI Name (Institution):	Uddin, Sardar Ph.D. (State University of New York, Stony Brook (Stony Brook University))		

Grant/Contract No.:	80NSSC19K0431
Performance Goal No.:	
Performance Goal Text:	
Task Description:	NOTE: Continuation of "Bone Remodeling under Differential Gravitational Environments--PI Uddin" with new Principal Investigator (PI) Dr. David Komatsu, who replaced previous PI, Dr. Sardar Uddin, in February 2020. Objective: Astronauts lose 1-1.5% of their bone mass per month during low Earth orbit missions. Spaceflight profiles of 2 years or longer, as projected for Mars missions, will significantly increase bone fragility, potentially compromising the ability of astronauts to perform their mission tasks in transit and on Mars. To facilitate human exploration of Mars, it is therefore essential to understand the underlying causes of microgravity-induced bone loss and develop effective countermeasures to sustain bone structure and mechanical integrity under microgravity conditions. Our long-term goal is to protect the skeletal health of astronauts in space environments so they can perform their mission operations and retain their quality of life upon return to Earth. The objective of the current proposal is to determine the effects of microgravity and partial microgravity on bone remodeling, as well as evaluate hypergravity as a countermeasure for microgravity-induced bone loss. Our central hypothesis is that simulated microgravity (Space 0 g) and partial microgravity (Mars 1/3 g, Moon 1/6 g) will impair bone remodeling and that hypergravity will mitigate microgravity-induced bone loss. The rationale for this proposal is the fact that microgravity and partial microgravity cause bone loss by reducing mechanical stresses on bone. Therefore, the application of periodic hypergravity will restore normal bone stresses, thereby rescuing physiological remodeling. Further study will also look at recovery rate in different microgravity conditions with and without periodic hypergravity exposure. Specific Aims and Methods: The central hypothesis will be tested by completing the following two specific aims. 1-Determine the effects of microgravity and partial microgravity on bone loss and long-term recovery using rat unloading models. Our working hypothesis is that microgravity (0 g) and partial microgravity (1/3 g and 1/6 g) will alter bone remodeling, leading to a long-term bone loss. Micro Computed Tomography (microCT) and histomorphometric analysis of bone formation and resorption will be conducted to determine microstructural changes. Mechanical testing will be used to assess structural integrity. The genomic effects of microgravity will be evaluated using RNA sequencing (RNA-seq) to identify differentially expressed signaling pathways. Longitudinal analysis of serum levels of bone markers will be conducted using enzyme-linked immunosorbent assay (ELISA) analysis. 2-Assess the efficacy of periodic hypergravity during simulated microgravity and partial microgravity as countermeasures to mitigate microgravity-induced short term and long term effects on bone integrity. Our working hypothesis is that hypergravity will provide essential mechanical stimulation to restore microgravity and partial microgravity-induced impairment of bone remodeling. The efficacy of periodic hypergravity will be determined using bone microstructural analyses (microCT, bone formation, and resorption histomorphometry), mechanical analysis (three-point testing), genomic (RNA-seq), and serum biomarker analysis (ELISA). Significance and Outcomes: Bone remodeling is a complex process that is tightly regulated by osteoblast-osteoclast coupling and communication. Microgravity impairs this process by increasing osteoclast associated resorption and decreasing osteoblastic bone formation. In order to continue to expand our crewed spaceflight program, it is essential to determine the effects of microgravity on bone remodeling and develop targeted countermeasures to sustain bone health in microgravity and partial microgravity. At the completion of the study, we expect to characterize the effects of microgravity and partial microgravity on bone structural and mechanical integrity, identify genomic effects of microgravity exposure, and establish the efficacy of periodic hypergravity to restore physiological bone remodeling. The outcomes of this study will fill NASA's recognized knowledge gaps (Osteo 4: We don't know the contribution of each risk factor on bone loss and recovery of bone strength, and which factors are the best targets for countermeasure application; Osteo 7: We need to identify options for mitigating early onset osteoporosis before, during, and after spaceflight); and research emphasis 3 (Microbiomes of the Built Environment (MoBE) of Spacecraft Element); and 4 (Molecular and Cellular Biology Element) (Appendix G).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	With the increasing expectancy of lifespan in the U.S. population, osteoporosis is becoming a more significant socio-economic burden on American society. There is a need to develop a therapeutic solution to osteopenia and osteoporosis. The data analyses of the proposed study will identify potential therapeutic targets to restore normal bone remodeling pathways. The analysis will focus on identifying both mechanical and molecular effects, and combine aspects of bone remodeling and mitigation of bone remodeling. The successful completion will provide us pro-anabolic and therapeutic targets to generate new bone in the osteopenic environment.
	Specific Aim 1: Identifying mechanistic changes in bone remodeling in microgravity, partial microgravity, and recovery phases. Live Animal Studies: The live animal studies proposed in Specific Aim 1 (SA1) have all been completed in full. The suspension system, including the vest used to hold the rats and the attachment mechanism to the top of the cage, were all modified in several preliminary runs to optimize rat movement and health. Bone Biomarker Evaluation: Serum from all rats was collected and is currently stored in our -80C freezer. Enzyme-linked immunosorbent assay (ELISA) kits have been ordered and are in storage. ELISA will be run this summer. Bone Quantity Analysis: To date, microcomputed tomography (microCT) analyses have been completed on 24 left femoral specimens. Of note, the microCT analyses have been delayed because the instrument initially expected to be used for the analyses went out of service and an alternative had to be found. This substantially delayed the scanning and subsequent analyses. The specimens remain blinded until the entire set have been scanned and analyzed. Bone Histomorphometric Analysis: All the left tibiae have been embedded in polymethyl methacrylate (PMMA), 32 have been sectioned and 20 have been imaged. Due to a software update, our previously established protocol is being modified to accommodate differences in workflow. Once completed, image analysis will proceed.

Task Progress:	Osteocyte Apoptosis and Activity: All the right tibiae have been embedded in paraffin and sectioned. Twenty samples (N=5) for the immediate sacrifice groups have undergone tartrate-resistant acid phosphatase (TRAP) staining and analysis. No differences were seen for regions of cortical bone; however, in the trabecular regions, there were significant increases in osteoclast surface, osteoclast surface/bone surface, osteoclast number, and osteoclast number/bone surface for the 0 G group compared to 1 G have been sectioned and 20 have been imaged. In addition, a significant increase in osteoclast number/bone surface was seen for 1/6 G compared to 1 G. Mechanical Testing: The left femora will undergo mechanical testing upon completion of the microCT analysis. Osteocyte Associated Mechanotransduction Pathway Analysis: RNA extraction has been completed for 32 right femora, corresponding to all of the specimens in the immediate recovery group. RNA sequencing (RNAseq) analysis has also been completed for this group. The experimental gravity groups were compared directly to the 1 G control group. For the differential expression analysis, the results showed that 1 G vs 1/3 G had the most differentially expressed genes, while 1G and 1/6 G had the most differentially expressed genes. This discrepancy suggests that partial gravity has more of an effect on global gene expression compared to microgravity. Gene ontology and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analysis of the differentially expressed genes is shown in heatmaps. The enrichment analysis showed that 1/6 G and 1/3 G have almost opposite patterns of downregulated or upregulated genes. Additionally, we noted that the 1 G group showed a large number of genes that are highly upregulated, where the 0 G group had many genes that were highly downregulated. Specific Aim 2: Assess the efficacy of periodic hypergravity during differential microgravity as a countermeasure to microgravity-induced short-term and long-term effects on bone integrity. Live Animal Studies: The live animal studies proposed in Specific Aim 2 (SA2) have all been completed in full. The same suspension system was used for suspension. In addition, for the hypergravity loading, lead weights were added to the bar at the top of the vest. Bone Biomarker Evaluation: Serum from all rats was collected and is currently stored in our -80C freezer. ELISA kits have been ordered and are in storage. ELISA will be run this summer. Bone Quantity Analysis: MicroCT analyses have not yet been conducted on any of the left femoral specimens. Bone Histomorphometric Analysis: All the left tibiae have been embedded in PMMA, none have yet been sectioned or analyzed. Osteocyte Apoptosis and Activity: All the right tibiae have been embedded in paraffin and sectioned. Apoptosis and TRAP staining will be completed in the future. Mechanical Testing: The left femora will undergo mechanical testing upon completion of the microCT analysis. Osteocyte Associated Mechanotransduction Pathway Analysis: RNA extraction has been completed for all of the right femora.
Bibliography Type:	Description: (Last Updated: 07/03/2024)
Abstracts for Journals and Proceedings	Meymarian E, Nihamul E, Divaris H, Farrell K, Uddin SMZ, Komatsu DE. "Down to the bone: Skeletal recovery after simulated Martian, lunar, and microgravity." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023